

FDS EXPERIMENT-11

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Experiment to understand pandas library using use cases in data science.

```
import numpy as np  
import pandas as pd
```

```
data = {  
    "Name": ["Arun", "Bala", "Charan", "Divya", "Ezhil"],  
    "Age": [20, 21, 19, 22, 20],  
    "Marks": [85, 90, 75, 95, 80]  
}  
  
df = pd.DataFrame(data)  
  
print(df)
```

```
...      Name  Age  Marks  
0     Arun   20    85  
1     Bala   21    90  
2  Charan   19    75  
3   Divya   22    95  
4   Ezhil   20    80
```

```
print(df.head(3))
print(df.tail(2))
print(df.describe())
```

```
...      Name  Age  Marks
0     Arun   20    85
1     Bala   21    90
2   Charan   19    75
      Name  Age  Marks
3   Divya   22    95
4   Ezhil   20    80

      Age  Marks
count  5.000000  5.000000
mean   20.400000  85.000000
std     1.140175   7.905694
min    19.000000  75.000000
25%    20.000000  80.000000
50%    20.000000  85.000000
75%    21.000000  90.000000
max    22.000000  95.000000
```

```
print("Student Names:")
print(df["Name"])

print("Students scoring above 80:")
print(df[df["Marks"] > 80])
```

```
Student Names:
0     Arun
1     Bala
2   Charan
3     Divya
4     Ezhil
Name: Name, dtype: object
Students scoring above 80:
      Name  Age  Marks
0     Arun   20    85
1     Bala   21    90
3   Divya   22    95
```

```
data = {  
    "Name": ["Arun", "Bala", "Charan", "Divya"],  
    "Marks": [85, None, 75, 95]  
}  
  
df = pd.DataFrame(data)  
  
print("Missing Values:")  
print(df.isnull().sum())  
  
df["Marks"] = df["Marks"].fillna(df["Marks"].mean())  
  
print("After Filling Missing Values:")  
print(df)
```

```
... Missing Values:  
Name      0  
Marks     1  
dtype: int64  
After Filling Missing Values:  
   Name  Marks  
0  Arun   85.0  
1  Bala   85.0  
2 Charan   75.0  
3  Divya   95.0
```

```
print("Total Marks:", df["Marks"].sum())
print("Average Marks:", df["Marks"].mean())
print("Maximum Marks:", df["Marks"].max())
print("Minimum Marks:", df["Marks"].min())
print("Number of Students:", df["Name"].count())
```

```
Total Marks: 340.0
Average Marks: 85.0
Maximum Marks: 95.0
Minimum Marks: 75.0
Number of Students: 4
```

```
data = {
    "Department": ["CSE", "ECE", "CSE", "ECE", "CSE"],
    "Marks": [85, 75, 90, 80, 95]
}

df = pd.DataFrame(data)

result = df.groupby("Department")["Marks"].mean()

print(result)
```

```
... Department
CSE      90.0
ECE      77.5
Name: Marks, dtype: float64
```